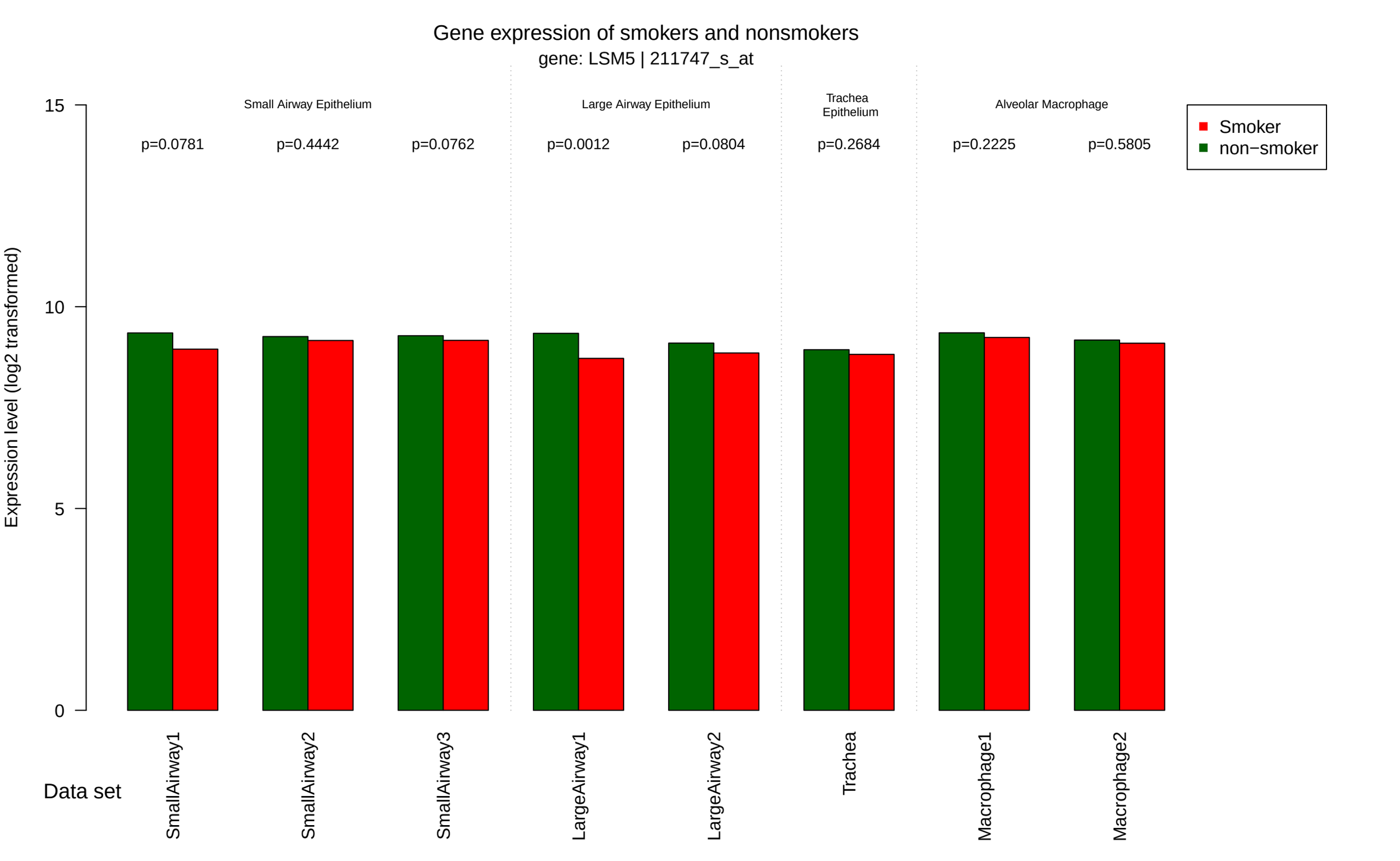
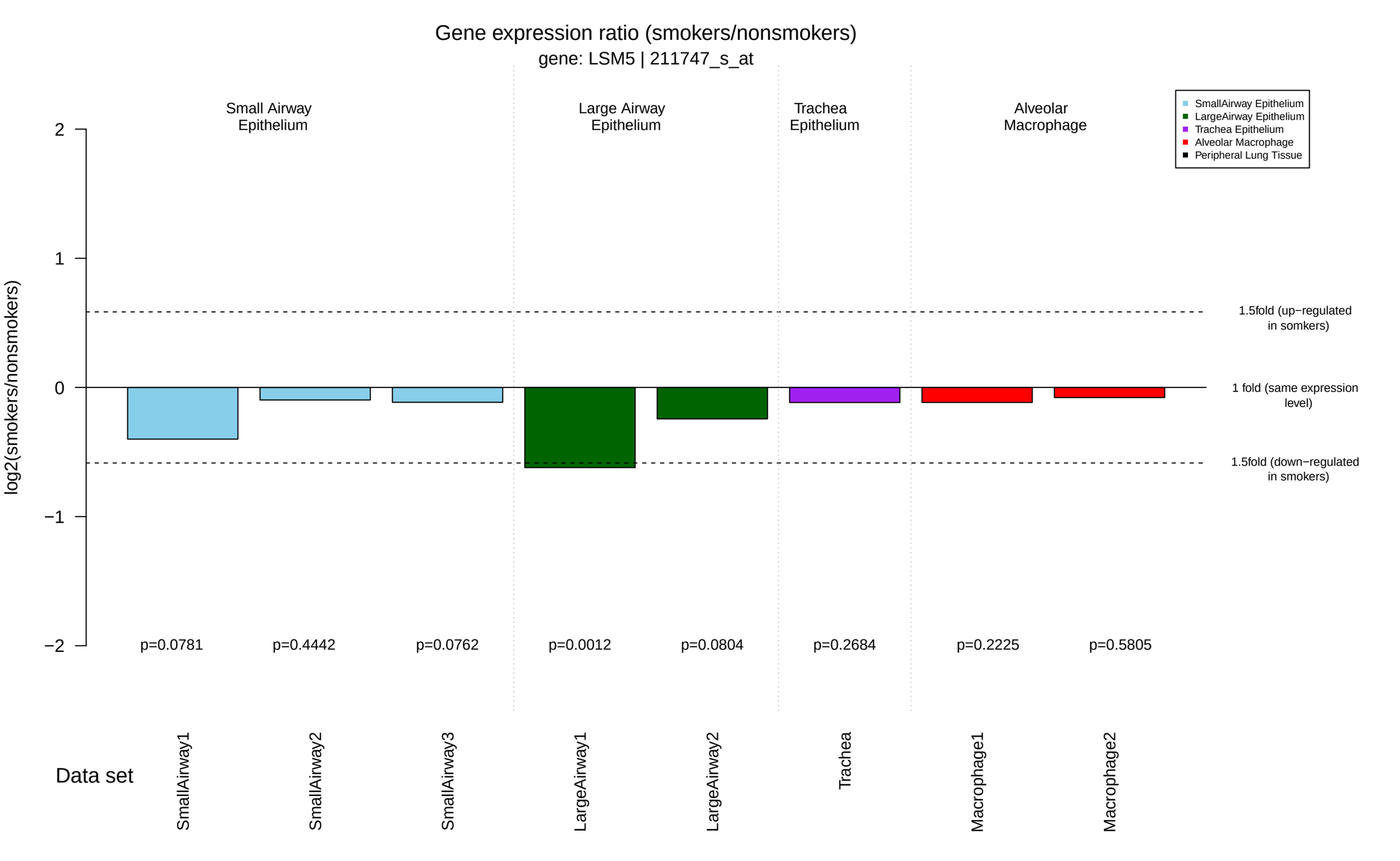
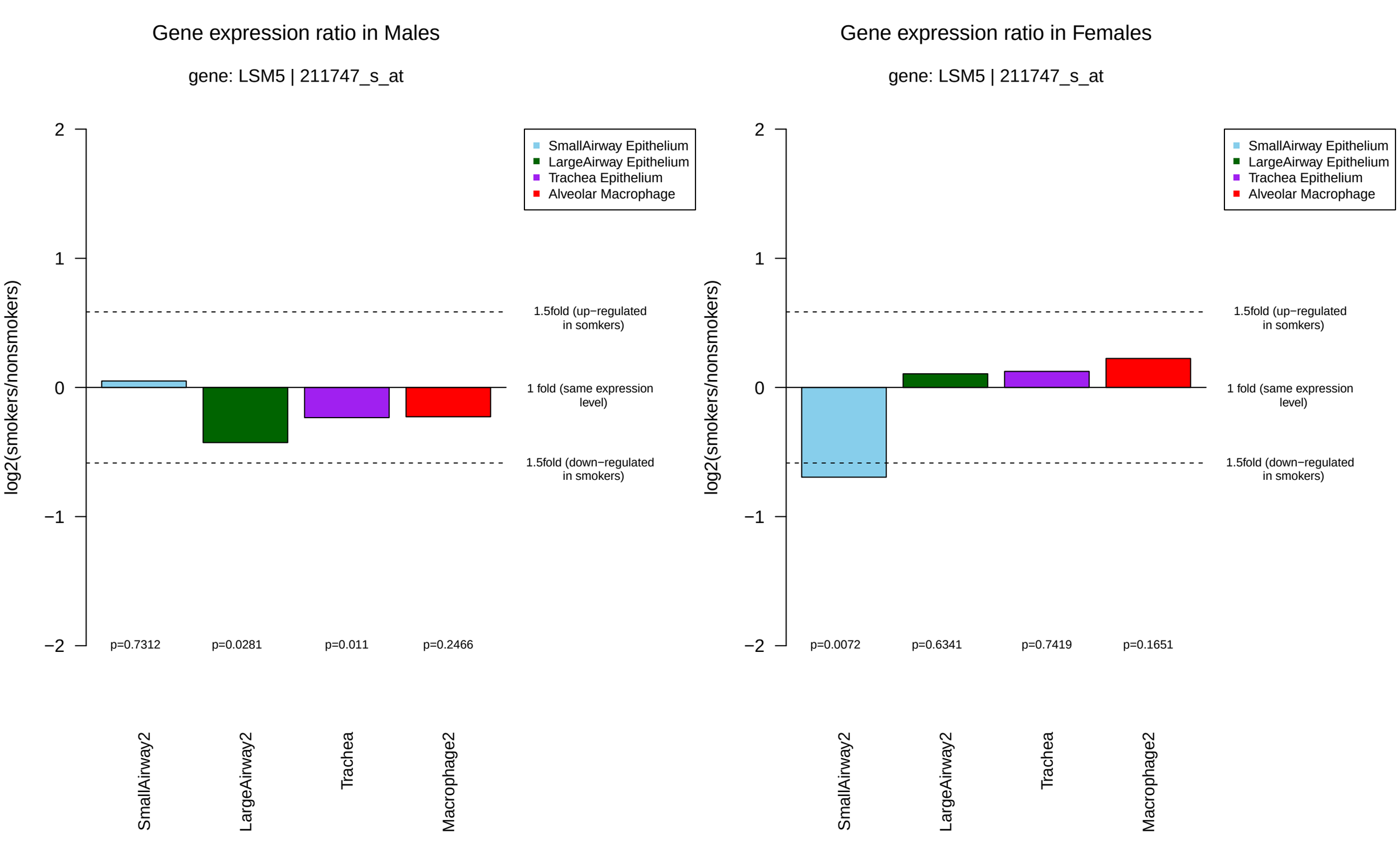
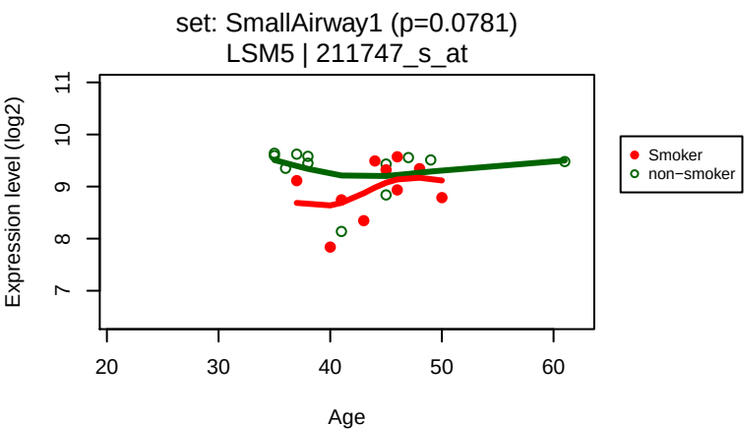
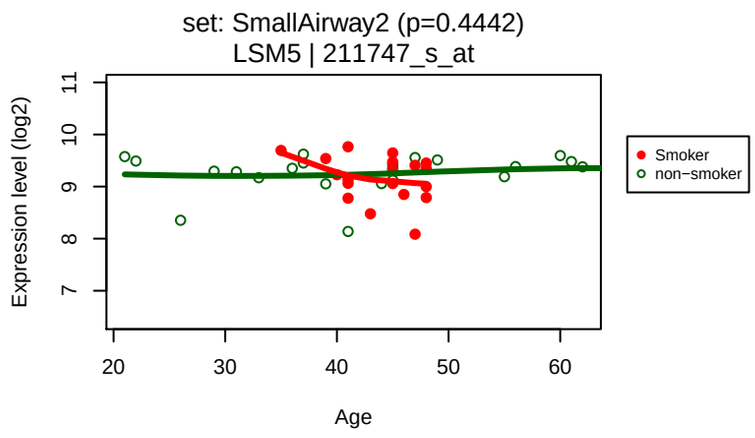
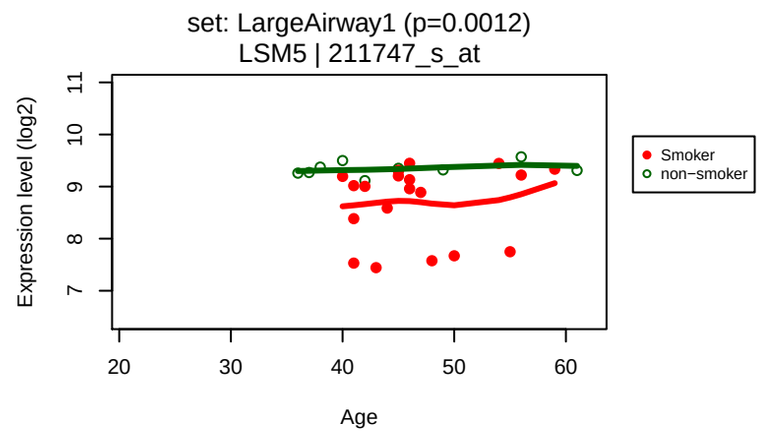

Gene expression of smokers and nonsmokers
gene: LSM5 | 211747_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0781
p=0.4442
p=0.0762
p=0.0012
p=0.0804
p=0.2684
p=0.2225
p=0.5805
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

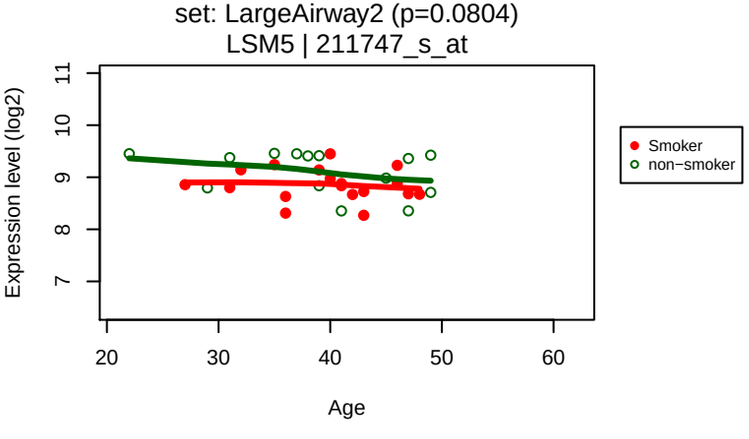

Gene expression ratio (smokers/nonsmokers)
gene: LSM5 | 211747_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0781
p=0.4442
p=0.0762
p=0.0012
p=0.0804
p=0.2684
p=0.2225
p=0.5805
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

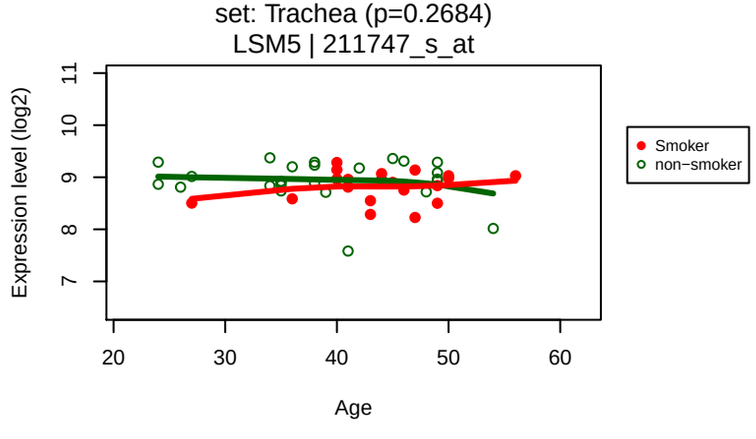

Gene expression ratio in Males
gene: LSM5 | 211747_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7312
p=0.0281
p=0.011
p=0.2466
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LSM5 | 211747_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0072
p=0.6341
p=0.7419
p=0.1651
SmallAirway2
LargeAirway2
Trachea
Macrophage2

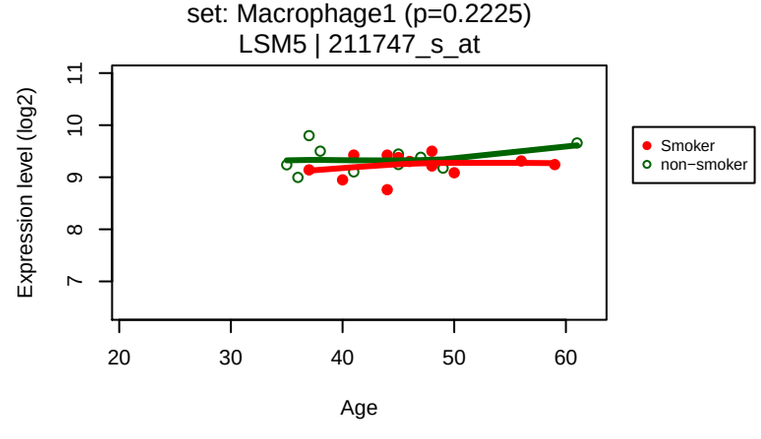

set: SmallAirway1 (p=0.0781)
LSM5 | 211747_s_at
Expression level (log2)
Age
Smoker
non−smoker

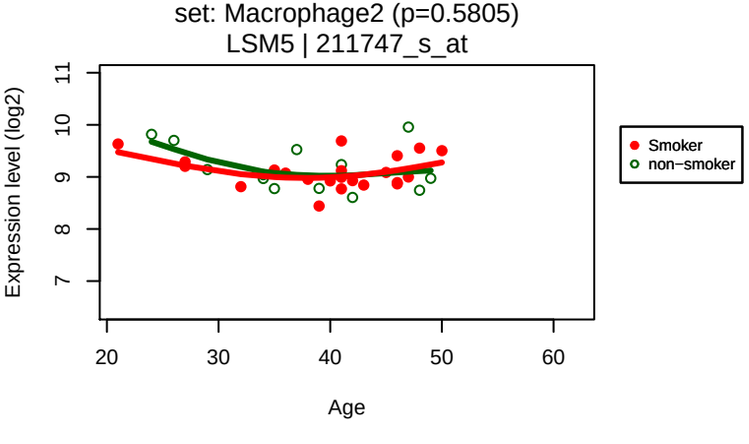

set: SmallAirway2 (p=0.4442)
LSM5 | 211747_s_at
Expression level (log2)
Age
Smoker
non−smoker

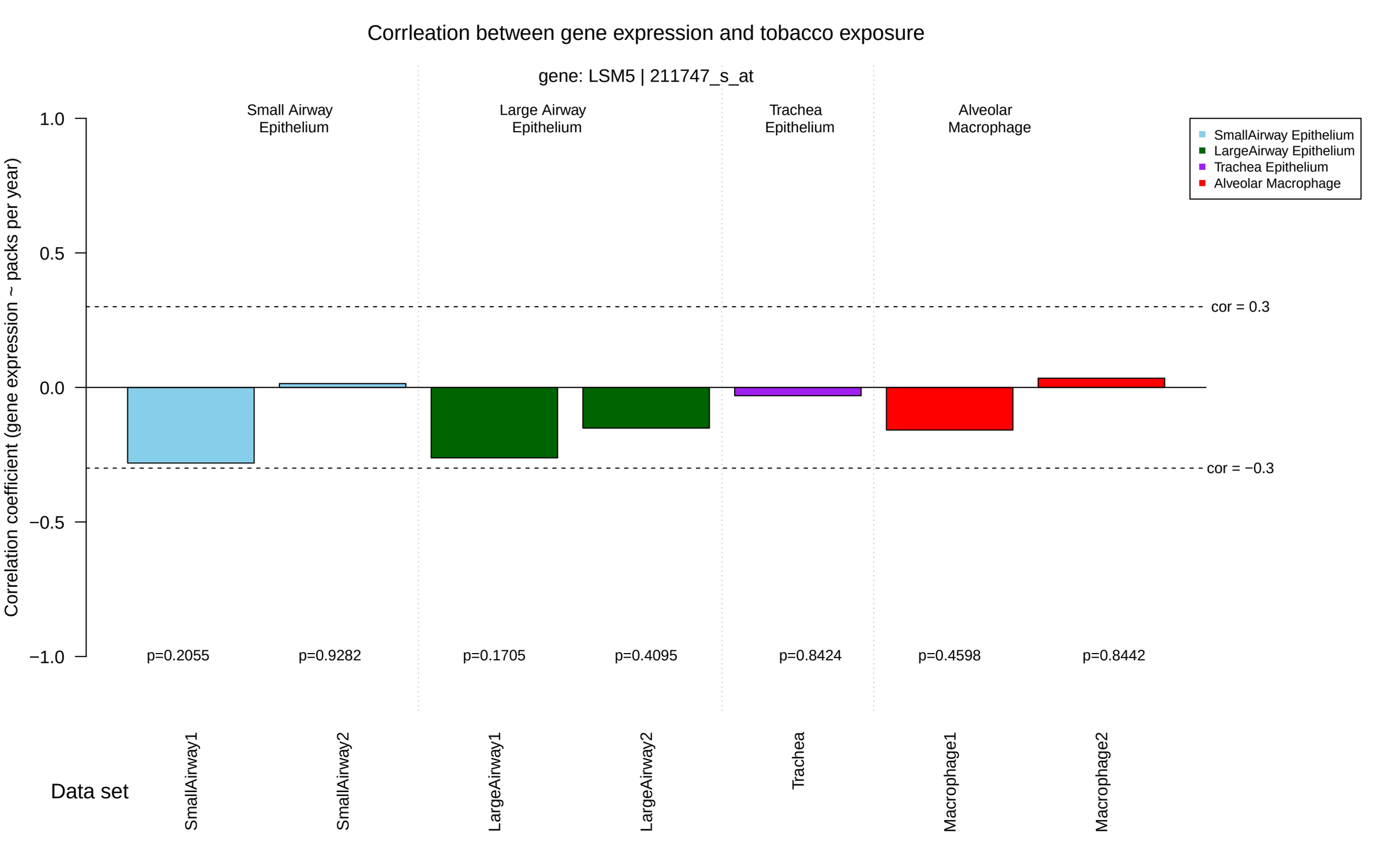

set: LargeAirway1 (p=0.0012)
LSM5 | 211747_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0804)
LSM5 | 211747_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2684)
LSM5 | 211747_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2225)
LSM5 | 211747_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5805)
LSM5 | 211747_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LSM5 | 211747_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2055
p=0.9282
p=0.1705
p=0.4095
p=0.8424
p=0.4598
p=0.8442
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2